

# Robot Learning Engineer Master Curriculum

Day 1 Packet - July 5, 2026

From Mechanical Engineering Student to Robotics + AI / Embodied AI Engineer

Today is not about finishing everything fast. Today is about building the system that will run for the next two years: coding environment, GitHub repository, daily practice, documentation, and honest progress reporting.

## Day Overview

| Block | Time       | Focus              | Main Output                               |
|-------|------------|--------------------|-------------------------------------------|
| 1     | 2 hours    | Python foundations | Basic Python programs committed to GitHub |
| 2     | 2 hours    | Git and GitHub     | Repository created and first commit made  |
| 3     | 1 hour     | DSA foundation     | 3 beginner problems solved and explained  |
| 4     | 20 minutes | Tool habit         | Git command notes                         |
| 5     | 10 minutes | Journal            | Daily progress entry written              |

# Primary Resources for Today

Use only the resources below today. Do not jump between random tutorials. Resource discipline is part of the training.

| Track                | Resource                                      | Use Today                                                                  |
|----------------------|-----------------------------------------------|----------------------------------------------------------------------------|
| Python course        | <a href="#">CS50P official course site</a>    | Open Week 0 / Lecture 0: Functions and Variables. Watch or read carefully. |
| Python lecture notes | <a href="#">CS50P Week 0</a>                  | Use while coding examples. Do not just passively watch.                    |
| Python documentation | <a href="#">Official Python tutorial</a>      | Read: Using the Python Interpreter and An Informal Introduction to Python. |
| Git book             | <a href="#">Pro Git official online book</a>  | Read Chapter 1 and Section 2.1 today.                                      |
| Algorithms course    | <a href="#">Princeton Algorithms Part I</a>   | Watch the course introduction only. Do not worry that examples use Java.   |
| Algorithms book      | <a href="#">Grokking Algorithms - Manning</a> | Read Introduction and Chapter 1 if you have legal access.                  |
| DSA practice         | <a href="#">LeetCode Array problem list</a>   | Solve the three beginner problems listed in this packet.                   |

Book note: Use legal access only. If a paid book is unavailable, use the official course/documentation and complete the code deliverables.

## Block 1 - Python Foundations

**Goal:** Write small command-line programs without copying.

- Open CS50P Week 0 / Lecture 0: Functions and Variables.
- Take notes on variables, input, print, strings, integers, floats, operators, functions, comments, and errors.
- Read the official Python tutorial sections on the interpreter and the informal introduction.
- Code everything inside a Day01 folder.

| File              | Program Requirement                                                            |
|-------------------|--------------------------------------------------------------------------------|
| hello.py          | Print Hello, world and one line about your goal.                               |
| intro.py          | Ask name, branch, college, and goal. Print a clean introduction.               |
| calculator.py     | Ask two numbers and print addition, subtraction, multiplication, and division. |
| temperature.py    | Convert Celsius to Fahrenheit and Fahrenheit to Celsius.                       |
| even_odd.py       | Ask an integer and print whether it is even or odd.                            |
| leap_year.py      | Check whether a year is a leap year.                                           |
| table.py          | Print multiplication table for a number from 1 to 10.                          |
| factorial.py      | Compute factorial using a loop.                                                |
| prime.py          | Check whether a number is prime.                                               |
| student_system.py | Mini project: collect student details and print a formatted summary.           |

```
robot-learning-journey/  
  Day01/  
    hello.py  
    intro.py  
    calculator.py  
    temperature.py  
    even_odd.py  
    leap_year.py  
    table.py  
    factorial.py  
    prime.py  
    student_system.py
```

## Block 2 - Git and GitHub

**Goal:** Start version control from the first day.

Create a GitHub repository named **robot-learning-journey**. Create README.md explaining your target role, focus areas, daily rule, and two-year goal.

```
git init
git status
git add .
git commit -m "Day 1: Python basics and Git setup"
git log --oneline
```

## Block 3 - DSA Foundation

**Goal:** Start problem solving slowly and correctly. Today is about explanation, not speed.

For each problem, write three lines in notes: idea, time complexity, and what mistake you made or avoided.

| Problem                 | Link                          | What to practice                                     |
|-------------------------|-------------------------------|------------------------------------------------------|
| Running Sum of 1D Array | <a href="#">LeetCode 1480</a> | Looping over a list and building cumulative sum.     |
| Richest Customer Wealth | <a href="#">LeetCode 1672</a> | Nested lists and maximum value.                      |
| Two Sum                 | <a href="#">LeetCode 1</a>    | Start with brute force. Do hash map later if needed. |

## Block 4 - Tool Habit

**Goal:** Spend 20 minutes only on Git habit. Do not over-study Git today.

| Command                 | Meaning                                     |
|-------------------------|---------------------------------------------|
| git init                | Start tracking a folder with Git.           |
| git status              | See which files changed and what is staged. |
| git add .               | Stage all current changes for commit.       |
| git commit -m "message" | Save a snapshot with a meaningful message.  |
| git log --oneline       | View commit history in compact form.        |

# Completion Checklist

- [ ] Development environment ready: Python, Git, VS Code, and terminal.
- [ ] GitHub repository created: robot-learning-journey.
- [ ] README.md written and committed.
- [ ] Day01 folder created with Python files.
- [ ] 10 Python programs completed.
- [ ] Student Information System mini project completed.
- [ ] Git commands practiced at least once.
- [ ] 3 DSA problems attempted and notes written.
- [ ] progress.md journal entry written.
- [ ] Final progress pasted into chat for Day 2 adjustment.

## What To Send Me After Finishing

Copy-paste this report into chat. This is how we adjust the next day without losing context.

Day 1 completed report:

1. Hours studied:
2. Python programs completed:
3. DSA problems completed:
4. GitHub link:
5. What I already knew:
6. What was difficult:
7. What I could not complete:
8. Questions:

## Quality Rules

- Do not copy code without understanding it yourself.
- Do not hide skipped tasks. Honest reporting makes the roadmap accurate.
- Every study day ends with code, notes, a GitHub commit, and a journal entry.
- Certificates are secondary. GitHub evidence, projects, research-quality work, and interview skill matter more.

## Source Notes and Resource Policy

Resources were chosen because they are official, widely respected, or highly practical for the current stage: CS50P, Python official documentation, Pro Git, Princeton Algorithms, Grokking Algorithms, and LeetCode. Resource discipline means finishing the assigned resource before opening random alternatives.